

SECTION A

1 (**short**) A thermally-insulated frictionless vertical piston with cross-sectional area $A = 0.01 \text{ m}^2$ contains a volume $V_1 = 0.01 \text{ m}^3$ of argon (modelled as a perfect gas with $\gamma = 5/3$) initially at temperature $T_1 = 300 \text{ K}$ and pressure $p_1 = 10^5 \text{ N m}^{-2}$.

- (a) A weight $F = 1000 \text{ N}$ is gently lowered onto the piston such that equilibrium is maintained at all times during the process. What is the pressure p_2 and volume V_2 of the argon at the end of this process? [2]
- (b) Calculate the change in internal energy, ΔU , of the argon during this process. [3]
- (c) The thermal insulation is removed and the argon returns to $T_3 = T_1 = 300 \text{ K}$. What is the final volume V_3 of the argon? [1]
- (d) How much heat is lost to the surroundings after the thermal insulation is removed? [3]
- (e) Explain why your answers to (b) and (d) are different. [1]